

Making reagents and buffers

- (9.) Once pH is adjusted, remove the stir bar and rinse the beaker walls into the solution with a small amount of reagent-grade water.

+ Optional: *Sterilizing buffers and solutions*

- (1.) Decide whether each solution is compatible with heat sterilization or requires filter sterilization. 

Safety: DO NOT AUTOCLAVE flammable liquids or solids such as ethanol, methanol, or chloroform; radioactive materials; chemicals that emit toxic or carcinogenic fumes when heated such as aldehydes or thiols; strong corrosives, including phenol, or bleach. Solutions that contain detergents can easily boil over.

- (2.) For autoclaving, loosely cap containers (one full turn) and wrap in autoclave paper or place in an autoclave tray. Run a liquid cycle for 20–60 min depending on container volume.

Critical: Steam-sterilize at 121 °C (250 °F) and 1.03–1.38 bar (15–20 psi). Allow slow exhaust to prevent boiling over. 

Quality assurance: Use autoclave indicator tape on each container to confirm exposure to steam. Tape alone does not guarantee sterility. For critical applications, include a chemical integrator strip inside the load or run periodic biological indicator tests with *Geobacillus stearothermophilus* spore ampules. 

- (3.) After cycle completion, wait 10–15 min for vapor removal before opening. Ensure that liquid remains clear and capped bottles are intact. Make up volume differences with autoclaved water. 
- (4.) *Optional:* Add supplements or adjust the pH of growth media or agar with sterile solutions when the autoclaved goods have cooled down.
- (5.) For filter sterilization, assemble a 0.2 µm pore size filter unit on a sterile funnel or manifold. Pre-wet the filter with sterile water if recommended by the manufacturer. 
- (6.) Slowly pour or draw the solution through the filter into a sterile bottle. Rinse the filter with 10 mL sterile buffer to maximize yield. 
- (7.) Label the sterilized solution with the word “sterile-filtered” or “autoclaved”.

>> *Storage of prepared buffers and solutions*

- (1.) Label each container clearly and persistently. Labels must be understood by everyone. Include:

- Name and concentration of the solution
- *Optional:* Recipe identifier
- Hazard labels (prominently, as applicable)
- Date (MM/YYYY) of preparation or expiry (as applicable)
- Your initials

- (2.) Protect light-sensitive solutions such as solutions containing certain antioxidants by wrapping containers in foil or storing in a dark cabinet, fridge, or freezer.

- (3.) *Optional:* For long-term storage (weeks to months) of frozen goods, aliquot into smaller volumes to reduce freeze-thaw cycles. Write the volume aliquoted on the secondary container.

Quality assurance: Aliquot buffers containing labile components such as EDTA-free protease inhibitors into single-use volumes. Aliquots stored refrigerated or frozen should occupy at least half of the container volume. 

- (4.) Store at the recommended temperature. 

Critical: Do not store stocks in a frost-free freezer! 

Making reagents and buffers

- (5.) Maintain a digital (or paper) inventory log of stored solutions and aliquoted materials  SOP0036. 
Record any deviations from the standard recipe, the date of preparation, when the container was opened, and when the batch was discarded.

 Troubleshooting (available online)  Notes (available online)

Lab Protocols — © B. C. Buchmuller. Licensed CC-BY-SA-4.0. Research use; provided as-is, without warranty. Current when printed. Visit <https://benjbuch.github.io/check/?q=c3d3f76> or scan the QR code to read the full protocol or to check for updates.

